Questions for the lecturer coaching

EAIF: AI for Finance, Group Assignment AS 2026 (ESG, stock returns and machine learning)

Group name: _____ **Date:** _____

Thank you for your time. We have kept the questions short; most of them state what we plan to do and ask whether that is acceptable. **How to use this sheet.** For each question, tick the box next to the answer the lecturer gives (the boxes can be ticked on screen or on paper) and write details in the notes field. The orange tag **our suggestion** marks the option we would pick if the lecturer has no preference. Questions tagged **PRIORITY** matter most if time is short: Q4, Q5, Q8, Q9 and Q11. Each question points to a section of the explainer document that explains the idea in plain terms.

Q1 Revenue unit of some firms

Explainer, section 1

About 22 to 27 firms (depending on the cut-off) report revenue and market cap roughly 1,000 times larger than the rest. Example: Company_ID 30 shows revenue 1,441,650 in 2012. In USD million that would be USD 1.4 trillion. Divided by 1,000 it is about 1,442, close to a typical firm (about 2,300 to 2,400). In our audit we plan to treat this as a unit mix-up and divide revenue and market cap by 1,000. Is any other treatment expected?

Why we ask: Audit (task a). We want to be sure that rescaling is an acceptable treatment.

- ☐ Plan is fine: treat as a unit mix-up and divide by 1,000 — ★ our suggestion
- ☐ Treat them as genuine large firms and keep the values
- ☐ Exclude these firms from the analysis
- ☐ Other: _____

NOTES (lecturer's answer)

Q2 Near-duplicate return rows

Explainer, section 2

300 company-months appear twice in the returns file with almost the same value. Example: Company_ID 1, March 2020: -27.7587 and -27.7549 (the two values differ by at most 0.01 percentage points). We plan to keep one row per company-month. Is that acceptable?

Why we ask: Audit (task a). If both rows stay, that month is counted twice when the 12-month return is computed.

- ☐ Plan is fine: keep one row per company-month (the first) — ★ our suggestion
- ☐ Replace the two rows by their average
- ☐ Drop both rows
- ☐ Other: _____

NOTES (lecturer's answer)

Q3 Return used as the target in tasks (b) and (c)

Explainer, section 3

We plan to forecast the total 12-month return in task (b), as in the decision-time rule, and to use the excess return (total return minus the risk-free rate) only in task (c). Is that the intended reading?

Why we ask: We only want to confirm. In the test years the two returns differ by about 5 points on average.

- [ ] Yes: total return in (b), excess return in (c) — ★ our suggestion
- [ ] Use the excess return in (b) as well
- [ ] Either is fine, as long as we state it clearly
- [ ] Other: _____

NOTES (lecturer's answer)

Q4 Formation year 2025 in the backtest

PRIORITY *Explainer, section 4*

Returns end in December 2025, so a portfolio formed in July 2025 has only 6 holding months. The starter's backtest includes this year, while its model fitting leaves it out. For task (d) we plan to keep formation years 2015 to 2025 in the backtest, to say in the memo that 2025 has only 6 months, and not to use 2025 for fitting. Is that acceptable, or should the last formation year be 2024?

Why we ask: Task (d) asks for a 12-month holding period. The backtest has 126 months with 2025 and 120 without.

- [ ] As described: 2025 in the backtest (6 months), not in model fitting — ★ our suggestion
- [ ] End at 2024: only portfolios with 12 holding months
- [ ] Other: _____

NOTES (lecturer's answer)

Q5 What does "before and from 2021" mean in task (c)?

PRIORITY *Explainer, section 5*

In task (c) we plan to define "before 2021" and "from 2021" by formation year, the July in which the portfolio is formed. A score for fiscal year 2020 is published in spring 2021, so it belongs to formation year 2021. Is that the intended reading?

Why we ask: The handout does not say which year defines the period. We want one definition throughout.

- [ ] Formation year (the score of fiscal year 2020 counts as "from 2021") — ★ our suggestion
- [ ] Fiscal year of the score (fiscal year 2020 counts as "before 2021")
- [ ] Calendar year in which the returns are earned
- [ ] Other: _____

NOTES (lecturer's answer)

Q6 Standardising the Environmental score in task (c)

Explainer, section 6

We plan to standardise the Environmental score within each year across all firms, as task (c) says, and to report the results by region. Is that right, or should it also be standardised within each region?

Why we ask: Standardising within region would change what one unit of the score means in each region.

- [ ] Within year across all firms — ★ our suggestion
- [ ] Within year and region
- [ ] Within year and industry
- [ ] Other: _____

NOTES (lecturer's answer)

Q7 A column with future information

Explainer, section 7

We found that the column Analyst_Outlook is strongly related to the NEXT 12-month return (rank correlation about 0.67) and to no other return window. We plan to exclude it from all models, report it in the audit table, and list it in the feature table as excluded. Is that documentation enough? (tick all that apply)

Why we ask: Audit (task a) and feature table. We want to document it in the expected way.

- [ ] Yes, that is enough — ★ our suggestion
- [ ] Also show a model with the column included, as a warning
- [ ] Other: _____

NOTES (lecturer's answer)

Q8 "Effect on one headline number" in the audit table

PRIORITY

Explainer, section 8

For "effect on one headline number" in the audit table, we plan to pick the number that fits each problem (for example mean return, test error or Sharpe ratio) and to say which one we used. Is that acceptable?

Why we ask: So that the audit table is complete and comparable. If specific numbers are prescribed, please write them in the notes.

- [ ] Free choice per problem, stating which number — ★ our suggestion
- [ ] One common headline number for all problems
- [ ] Other: _____

NOTES (lecturer's answer)

Q9 Page and slide limits

PRIORITY *Explainer, section 9*

(a) We assume that tables and figures count towards the 3 memo pages and that only the appendix is extra. Is that correct? (b) Does the limit of 8 slides include the title slide?

Why we ask: Rules only. We need to plan the layout of the memo and the slides.

(a) Memo, 3 pages plus appendix

- [ ] Yes: everything counts, only the appendix is extra — our suggestion
- [ ] No: only the text counts; tables and figures may go to the appendix

(b) Slides, at most 8

- [ ] Title slide is included in the 8 — our suggestion
- [ ] Title slide is extra
- [ ] Other: _____

NOTES (lecturer's answer)

Q10 Libraries and run time in Colab

Explainer, section 10

We plan to use scikit-learn, statsmodels (regressions with standard errors) and possibly lightgbm, all of which run in Colab. Is that acceptable? Is there a limit on how long the notebook may run?

Why we ask: Tasks (b), (c), (d). The notebook must run top to bottom in Colab without errors.

- [ ] Plan is fine — our suggestion
- [ ] Only libraries that are already installed in Colab
- [ ] Only scikit-learn and what the starter imports

Maximum run time of the notebook (minutes): _____

- [ ] Other: _____

NOTES (lecturer's answer)

Q11 Use of AI tools

PRIORITY *Explainer, section 11*

If we use AI assistants (for example for coding or drafting), is that allowed for this assignment, and how should we declare it?

Why we ask: This is the lecturer's rule, so we have no suggestion and will follow the answer.

- ☐ Allowed, declared in the memo
- ☐ Allowed, with a separate declaration (tools used, prompts, or a form)
- ☐ Allowed, no declaration needed
- ☐ Not allowed

How and where to declare: _____

☐ Other: _____

NOTES (lecturer's answer)

Q12 Is our notebook the final version?

Explainer, section 12

Our copy of Group_Work_AS_2026.ipynb still says "DEADLINE: <to be set by the instructor>". Is this the final version of the notebook, and where do we get updates?

Why we ask: Rules only. We do not want to build on a version that is replaced later.

- ☐ Yes, it is final
- ☐ A newer version exists (write link or place below)

Link or place: _____

☐ Other: _____

NOTES (lecturer's answer)

Our own questions (space for more)

Q13

Q14

Q15